

HabitFlow API Specification

Team-404

Version 1.1

Last Updated: 2026, Sept. 30

Prepared by: Eunice Grace O. Revillas (Task Scheduling API Spec)

Endpoints marked with an Authorization parameter require the header Authorization: Bearer <token> from /api/auth/login. Every authenticated request only sees data owned by that user.

IV. Task Scheduling (api/schedule, api/tasks)

A. Scheduling Rules

Scheduling uses the dueDate, dueTime, deadline and completedAt fields of a task. Status is computed in the user's timezone (set at signup, default Asia/Manila, changeable through Update Profile I.F). Dates use YYYY-MM-DD and times use HH:mm:ss.

Status	Definition
unscheduled	Not completed and dueDate is null.
current	Not completed and dueDate is today.
upcoming	Not completed and dueDate is after today.
overdue	Not completed and dueDate (plus dueTime, if set) is earlier than now.
completed	completedAt is not null.

- dueTime requires dueDate. deadline must not be earlier than dueDate.
- A task that matches both current and overdue (due today at a time that has passed) is reported as overdue.
- These rules apply to every endpoint that changes scheduling fields: Add Task (III.C), Update Task Due Date / Due Time / Deadline (III.J to III.L) and Reschedule Task (IV.C).
- Archived tasks (III.N) are excluded from List Tasks by Schedule (IV.B) and from the Task Dashboard counts (VI.A).
- The Task Dashboard counts (currentCount, upcomingCount, overdueCount, completedCount) use these same definitions.

Recurring tasks use a single task row. When a recurring task is completed, the server increments the streak (if tracked), moves dueDate (and deadline, by the same offset) to the next occurrence and keeps completedAt null, so the task stays active. If the next occurrence falls after the rule's endDate, the task is completed normally instead. If an occurrence becomes overdue without being completed, currentStreak resets to 0.

daysOfWeek is sent and returned as an array of integers (1 = Mon to 7 = Sun). It is stored in days_of_week (SMALLINT) as a bitmask: Mon = 1, Tue = 2, Wed = 4, Thu = 8, Fri = 16, Sat = 32, Sun = 64. For example, [1, 3, 5] is stored as 21.

B. List Tasks by Schedule

API Endpoint	/api/schedule/tasks
method	GET

Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
status	string	Optional. current, upcoming, overdue, completed, or unscheduled. Omit for all.
from	string	Optional. Start date (YYYY-MM-DD), matched against dueDate.
to	string	Optional. End date (YYYY-MM-DD), matched against dueDate.
<i>Lists the authenticated user's active (non-archived) tasks filtered by schedule, ordered by dueDate (earliest first). Filters are query parameters and can be combined.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
[]	array	List of tasks with the same fields as List Tasks (III.A).
[]taskId	integer	Task ID.
[]projectId	integer	Project ID.
[]priorityId	integer	Priority ID.
[]title	string	Task title.
[]description	string/null	Task description.
[]dueDate	string/null	Due date.
[]dueTime	string/null	Due time.
[]deadline	string/null	Deadline.
[]completedAt	string/null	Completion timestamp.
[]sortOrder	integer	Display order within the project.
[]isRecurring	boolean	True if the task has a recurrence rule.
[]isArchived	boolean	Always false in this list.
[]labels	array	Labels attached to the task (same shape as III.A).
[]status	string	Computed status (see IV.A).
Sample Response		
<pre>[{"taskId": 1, "projectId": 10, "priorityId": 4, "title": "Study ILT Module 4", "description": "Read all lessons in the module and finish the code", "dueDate": "2026-10-02", "dueTime": "18:00:00", "deadline": "2026-10-02", "completedAt": null, "sortOrder": 0, "isRecurring": false, "isArchived": false, "labels": [{"labelId": 3, "name": "Study", "color": "#2563EB"}], "status": "upcoming"}]</pre>		
Fail		
Status Code	400	

Response Type	json	
Response Data	Type	Description
error	string	Invalid filter or date range.
Sample Response		
{"error": "Invalid date range"}		
Fail		
Status Code	401	
Response Type	json	
Response Data	Type	Description
error	string	Authentication error.
Sample Response		
{"error": "Unauthorized"}		

C. Reschedule Task

API Endpoint	/api/tasks/{taskId}/schedule	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
taskId	integer	Existing task ID (path parameter).
dueDate	string/null	New due date. null clears the schedule (and dueTime).
dueTime	string/null	New due time. Requires dueDate.
deadline	string/null	New deadline. Must not be earlier than dueDate.
<i>Updates several scheduling fields of a task at once (e.g. from a date picker). Fields not sent are left unchanged. To change a single field, use Update Task Due Date, Due Time or Deadline (III.J to III.L). A recurring task's dueDate cannot be cleared.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
taskId	integer	Task ID.
dueDate	string/null	Due date.
dueTime	string/null	Due time.
deadline	string/null	Deadline.
status	string	Computed status after the change.
message	string	Success confirmation.

Sample Response		
{ "taskId": 25, "dueDate": "2026-10-05", "dueTime": "09:00:00", "deadline": "2026-10-07", "status": "upcoming", "message": "Task rescheduled successfully" }		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Invalid date or time.
Sample Response		
{ "error": "Deadline cannot be earlier than due date" }		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Task not found or not accessible.
Sample Response		
{ "error": "Task not found" }		

D. Complete / Reopen Task

API Endpoint	/api/tasks/{taskId}/complete	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
taskId	integer	Existing task ID (path parameter).
completed	boolean	true completes the task. false clears completedAt (reopens it).
<i>The only endpoint that changes completion. For a non-recurring task, completed=true sets completedAt to the current time. For a recurring task, it advances the task to its next occurrence as described in IV.A. Completed tasks are counted in completedCount on the Task Dashboard.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
taskId	integer	Task ID.
completedAt	string/null	Completion timestamp (null for a recurring task that advanced).

dueDate	string/null	Due date after the change (the next occurrence for a recurring task).
currentStreak	integer/null	Current streak (recurring tasks with trackStreak only).
message	string	Success confirmation.
Sample Response		
{ "taskId": 25, "completedAt": null, "dueDate": "2026-10-07", "currentStreak": 4, "message": "Task completed successfully" }		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Task not found or not accessible.
Sample Response		
{ "error": "Task not found" }		

E. Set Recurrence

API Endpoint	/api/tasks/{taskId}/recurrence	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
taskId	integer	Existing task ID (path parameter).
frequency	string	Required. daily, weekly, or monthly.
interval	integer	Repeat every N units (default is 1, min is 1).
daysOfWeek	array of integer	Weekly only. 1 (Mon) to 7 (Sun). Defaults to the weekday of dueDate. Stored as a bitmask (see IV.A).
dayOfMonth	integer/null	Monthly only. 1 to 31. Defaults to the day of dueDate.
endDate	string/null	Optional last date an occurrence may fall on.
trackStreak	boolean	Track completion streaks (default is false).
<i>Creates or replaces the recurrence rule of a task. A task has at most one rule, and the task must have a dueDate. Existing streak counts are kept when the rule is replaced.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
taskId	integer	Task ID.

frequency	string	daily, weekly, or monthly.
interval	integer	Repeat every N units (interval_count in the database).
daysOfWeek	array of integer/null	Weekly only. 1 (Mon) to 7 (Sun).
dayOfMonth	integer/null	Monthly only. 1 to 31.
endDate	string/null	Last date an occurrence may fall on.
trackStreak	boolean	Whether completion streaks are tracked.
currentStreak	integer	Current streak count.
longestStreak	integer	Longest streak count.
message	string	Success confirmation.
Sample Response		
{"taskId": 25, "frequency": "weekly", "interval": 1, "daysOfWeek": [1, 3, 5], "dayOfMonth": null, "endDate": "2026-12-31", "trackStreak": true, "currentStreak": 3, "longestStreak": 7, "message": "Recurrence set successfully"}		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.
Sample Response		
{"error": "Task must have a due date"}		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Task not found or not accessible.
Sample Response		
{"error": "Task not found"}		

F. View Recurrence

API Endpoint	/api/tasks/{taskId}/recurrence	
method	GET	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
taskId	integer	Existing task ID (path parameter).
<i>Returns the recurrence rule and streak counts of a task.</i>		

Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
taskId	integer	Task ID.
frequency	string	daily, weekly, or monthly.
interval	integer	Repeat every N units (interval_count in the database).
daysOfWeek	array of integer/null	Weekly only. 1 (Mon) to 7 (Sun).
dayOfMonth	integer/null	Monthly only. 1 to 31.
endDate	string/null	Last date an occurrence may fall on.
trackStreak	boolean	Whether completion streaks are tracked.
currentStreak	integer	Current streak count.
longestStreak	integer	Longest streak count.
Sample Response		
<pre>{ "taskId": 25, "frequency": "weekly", "interval": 1, "daysOfWeek": [1, 3, 5], "dayOfMonth": null, "endDate": "2026-12-31", "trackStreak": true, "currentStreak": 3, "longestStreak": 7 }</pre>		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Task not found, not accessible, or has no recurrence rule.
Sample Response		
<pre>{ "error": "Recurrence not found" }</pre>		

G. Remove Recurrence

API Endpoint	/api/tasks/{taskId}/recurrence	
method	DELETE	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
taskId	integer	Existing task ID (path parameter).
<i>Removes the recurrence rule (and its streak counts) from a task. The task stays as a normal task with its current dueDate.</i>		
Success		

Status Code	200	
Response Type	json	
Response Data	Type	Description
message	string	Recurrence removal confirmation.
Sample Response		
{ "message": "Recurrence removed successfully" }		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Task not found, not accessible, or has no recurrence rule.
Sample Response		
{ "error": "Recurrence not found" }		